

1.Introduction Activation and Synaptic Dynamics

Artificial Neural Networks (ANNs) are mathematical models inspired by the structure and functioning of biological neural systems. They consist of interconnected neurons that process information through weighted connections. The intelligent behaviour of these networks depends mainly on two fundamental mechanisms: activation dynamics and synaptic dynamics.

Activation Dynamics

Activation dynamics describe how a neuron's internal state and output evolve in response to input signals. When a neuron receives inputs, each input is multiplied by a corresponding weight, summed together, and combined with a bias term. This combined signal determines the neuron's activation level.

The net input to a neuron is given by:

$$u = \sum_{i=1}^n w_i x_i + b$$

The output is then determined by:

$$y = f(u)$$

In dynamic systems, activation can also depend on time:

$$\tau \frac{du(t)}{dt} = -u(t) + \sum w_i x_i(t)$$

Here:

- $u(t)$ represents the neuron's internal state (membrane potential),
- τ is a time constant controlling response speed.

Activation dynamics control how information flows through the network and determine the network's response to inputs.

Synaptic Dynamics

Synaptic dynamics describe how the connection weights between neurons change during learning. In biological systems, synapses strengthen or weaken based on experience.

A general weight update rule is:

$$\Delta w = -\eta \frac{\partial E}{\partial w}$$

Where:

- E is the error (loss) function,
- η is the learning rate,

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Where:

- E is the error (loss) function,
- η is the learning rate,
- $\frac{\partial E}{\partial w}$ is the gradient of error with respect to weight.

Activation Dynamics Models

Activation dynamics describe how the internal state of a neuron evolves and produces an output in response to input signals. Unlike simply defining an activation function, activation dynamics focus on how neuron activity changes over time or across layers during signal propagation.

Activation dynamics determine:

- How inputs are integrated
- How neuron states evolve
- How outputs are generated
- How information flows through the network

Basic Static Activation Model

In most feedforward neural networks, activation is computed in two steps:

Step 1: Net Input Calculation

$$u = \sum_{i=1}^n w_i x_i + b$$

Where:-

- x_i = Inputs
- w_i = Weights

- b = Bias
- u = Internal state (membrane potential)

Step 2: Output Generation

$$y=f(u)$$

Here, $f(u)$ determines how strongly the neuron activates.

This represents **instantaneous activation dynamics**, where output depends only on current input.

Continuous-Time Activation Dynamics

Used in dynamic or biologically inspired models:

$$\tau \frac{du(t)}{dt} = -u(t) + \sum_{i=1}^n w_i x_i(t) + b$$

Where:

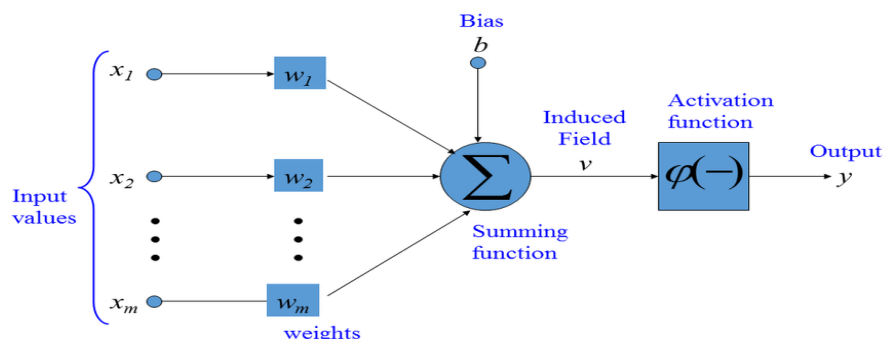
- $u(t)$ = Time-dependent activation
- τ = Time constant
- $du(t)$ = Rate of change of activation

Output:

$$y(t)=f(u(t))$$

This equation includes:

- Decay term: $-u(t)$
- Excitation from inputs
- Time evolution of activation



Discrete-Time Activation Dynamics

Common in artificial neural networks:

$$u^{(k)} = Wx^{(k)} + b$$

$$y^{(k)} = f(u^{(k)})$$

Where:

- k = Iteration or layer index

Recurrent Activation Dynamics (RNN)

Activation depends on previous state:

$$h_t = f(W_{hh}h_{t-1} + W_{xh}x_t + b)$$

Where:

- h_t = Current hidden state
- h_{t-1} = Previous hidden state
- x_t = Current input
- Hopfield network.

Shunting Activation Models

Shunting activation models are a type of **nonlinear dynamic neuron model** used to describe realistic neural behaviour. These models are widely used in biologically inspired neural networks because they incorporate bounded activation and multiplicative interactions between excitation and inhibition.

Unlike simple additive models, shunting models normalize the neuron response automatically and prevent unlimited growth of activation.

Basic Shunting Activation Equation

The general form of a shunting neuron model is:

$$\frac{du}{dt} = -Au + (B - u)E - (u - C)I$$

Where:

- u = Neuron activation (membrane potential)
- A = Passive decay rate
- B = Upper bound of activation
- C = Lower bound of activation
- E = Excitatory input
- I = Inhibitory input

Explanation of Terms

1. Decay Term

$$-Au$$

Represents passive leakage of activation.

2. Excitatory Shunting Term

$$(B-u)E$$

- Excitation decreases as activation approaches upper bound BBB.
- Prevents unlimited growth.

3. Inhibitory Shunting Term

$$-(u-C)I$$

- Inhibition decreases as activation approaches lower bound CCC.
- Prevents activation going below lower bound.

Bounded Activation Property

The model guarantees:

$$C \leq u(t) \leq B$$

This ensures stability and biological realism.

Simplified Shunting Model

Sometimes written as:

$$\frac{du}{dt} = -u + (1-u)E - uI$$

- Upper bound $B=1$
- Lower bound $C=0$
- Decay constant normalized

Advantages of Shunting Models

- Automatic normalization
- Bounded activation
- Stable behaviour
- Realistic biological modeling
- Used in vision and pattern recognition models

Applications

- Competitive neural networks
- Adaptive resonance theory (ART)
- Visual cortex modelling.
- Image processing systems

Shunting activation models provide a more biologically realistic and mathematically stable form of activation dynamics. By incorporating multiplicative excitation and inhibition, they ensure bounded and normalized neuron responses, making them useful in dynamic neural systems.

Activation dynamics define how neuron states evolve and how outputs are produced in response to inputs. They regulate signal processing within the network and form the foundation of forward propagation. Understanding activation dynamics is essential for analyzing neural network stability, efficiency, and performance.

Synaptic Dynamics – Important Points

Basic Concept

- Synaptic dynamics refers to the process of updating connection weights between neurons.
- It represents learning in neural networks.
- Weights store knowledge in the network.
- Learning occurs by modifying weights based on input and error.

Mathematical Foundation

- Learning is based on minimizing a loss function $E(W)$.
- General update rule:

$$\Delta w = -\eta \frac{\partial E}{\partial w}$$

- Gradient indicates direction of steepest error increase.
- Weight update moves in opposite direction to reduce error.

Types of Synaptic Learning Rules

1. Supervised Learning

- Uses target output.
- Example: Delta rule, Backpropagation.
- Minimizes error directly.

2. Unsupervised Learning

- No target output.
- Based on correlation.
- Example: Hebbian learning.

3. Competitive Learning

- Only one neuron updates weights.
- Used in clustering.

Properties of Synaptic Dynamics

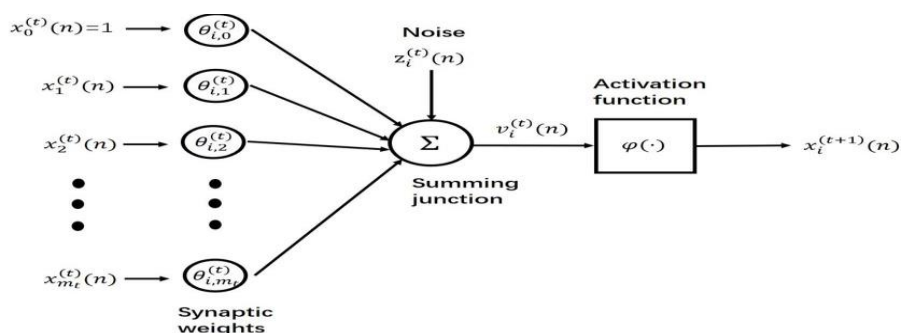
- Adaptive in nature.
- Iterative process.
- Depends on learning rate.
- May converge or diverge.
- Can be discrete-time or continuous-time.
- Sensitive to initial weights.

Stability Conditions

- Small learning rate $\rightarrow$ Stable but slow learning.
- Large learning rate $\rightarrow$ Fast but unstable.
- Convergence when gradient approaches zero.
- Stability often analyzed using eigenvalues.

Role in Backpropagation

- Error is propagated backward.
- Each layer adjusts weights.
- Hidden layers learn internal representations.
- Enables deep learning.



Biological Interpretation

- Based on synaptic plasticity.
- Strengthening = Long-Term Potentiation (LTP).
- Weakening = Long-Term Depression (LTD).
- Inspired by “neurons that fire together wire together.”

Advantages

- Enables learning from data.
- Improves prediction accuracy.
- Adapts to new patterns.
- Stores information efficiently.

Limitations

- Can get stuck in local minima.
- Slow convergence in deep networks.
- Sensitive to hyperparameters.
- Risk of overfitting.

Conclusion

The equations of **Activation Dynamics** and **Synaptic Dynamics** form the mathematical foundation of neural networks. Activation dynamics describe how neurons process input signals and generate outputs, while synaptic dynamics define how connection weights are adjusted during learning. Together, they govern both the forward flow of information and the adaptive learning process within the network.

Activation equations ensure non-linearity, stability, and proper signal transformation, enabling neural networks to model complex real-world patterns. Synaptic update equations, derived from optimization principles such as gradient descent and error minimization, allow the network to iteratively reduce error and improve performance over time.

Aspect	Activation Dynamics	Synaptic Dynamics
Definition	Describes how neuron states evolve and produce outputs	Describes how connection weights change during learning
Purpose	Signal processing (forward propagation)	Learning and adaptation (weight update)
Main Variable	Neuron activation (u, y)	Synaptic weight (w_{ij})
Basic Equation	$u = \sum_{i=1}^n w_i x_i + b$	$\Delta w = -\eta \frac{\partial E}{\partial w}$
Time Dependency	May be static or time-dependent ($\frac{du(t)}{dt}$)	Often iterative or continuous ($\frac{\partial E}{\partial w}$)
Role in Network	Determines output response	Minimizes error and improves performance

Aspect	Activation Dynamics	Synaptic Dynamics
Mathematical Nature	Nonlinear transformation ($y = f(u)$)	Optimization / gradient-based adjustment
Occurs During	Forward pass	Backward pass (training phase)
Influences	Signal strength, non-linearity, stability	Convergence, learning speed, accuracy
Biological Inspiration	Membrane potential and firing behaviour	Synaptic plasticity (LTP & LTD)
Examples	ReLU, Sigmoid, Tanh dynamics	Hebbian rule, Delta rule, Backpropagation
Failure Without It	Network becomes linear and weak	Network cannot learn

The interaction between these two mechanisms enables neural networks to learn from data, adapt to changing inputs, and achieve convergence toward optimal solutions. Without activation dynamics, networks would lack expressive power; without synaptic dynamics, they would lack learning capability.

In summary, the equations of activation and synaptic dynamics are essential for understanding how neural networks function, learn, and achieve intelligent behaviour in applications such as image recognition, speech processing, time-series prediction, and artificial intelligence systems.

Activation Dynamics – Signal Processing

Activation dynamics describe how a neuron transforms inputs into outputs and how its internal state evolves.

The basic activation equation is:

$$u = \sum_{i=1}^n w_i x_i + b$$

For dynamic systems (continuous-time model):

$$\tau \frac{du(t)}{dt} = -u(t) + \sum w_i x_i(t)$$

These equations show that:

- The neuron integrates weighted inputs.
- Activation may decay over time.
- The activation function introduces non-linearity.

Thus, activation dynamics control the **forward propagation of signals** in the network.

Synaptic Dynamics – Learning Mechanism

Synaptic dynamics describe how weights are updated to minimize error and improve performance.

The general learning rule based on gradient descent is:

$$\Delta w_{ij} = -\eta \frac{\partial E}{\partial w_{ij}}$$

For Mean Squared Error:

$$E = \frac{1}{2}(t - y)^2$$

These equations ensure:

- Error reduction over iterations.
- Convergence toward optimal weight value.